13. Classes and Objects

In C++, classes and objects are utilized to enable Object Oriented Programming. A class is a custom-defined data type that acts as a model for an object. It comprises data members, also known as attributes, and member methods, which operate on that data. Objects, on the other hand, are instances of a class that are necessary to utilize class data and methods since they cannot be accessed directly.

14. Decision Making

In programming, there are situations where we need to make decisions and based on these decisions, we determine what actions to take next. These decision-making situations can be handled using statements in programming languages that determine the flow of program execution.

15. Forward declarations in C++

Forward declaration in programming refers to the act of declaring the syntax or signature of an identifier, variable, function, class, etc. before its usage in the program. In C++, forward declaration is commonly used for classes, where the class is defined before its use so that it can be called and used by other classes that are defined earlier in the program.

16. Errors in C++

Programming errors, commonly known as bugs, occur when the user performs an illegal operation that causes abnormal program behavior. These errors can remain unnoticed until the program is compiled or executed. Certain errors can even prevent the program from compiling or executing altogether. Therefore, it is essential to identify and remove errors before compiling and executing the program.

17. Undefined behaviours

If a beginner in C/C++ programming is not familiar with the concept of undefined behavior, it may cause various problems in the future. It can make debugging difficult, especially when trying to trace the root cause of an undefined error in someone else's code.